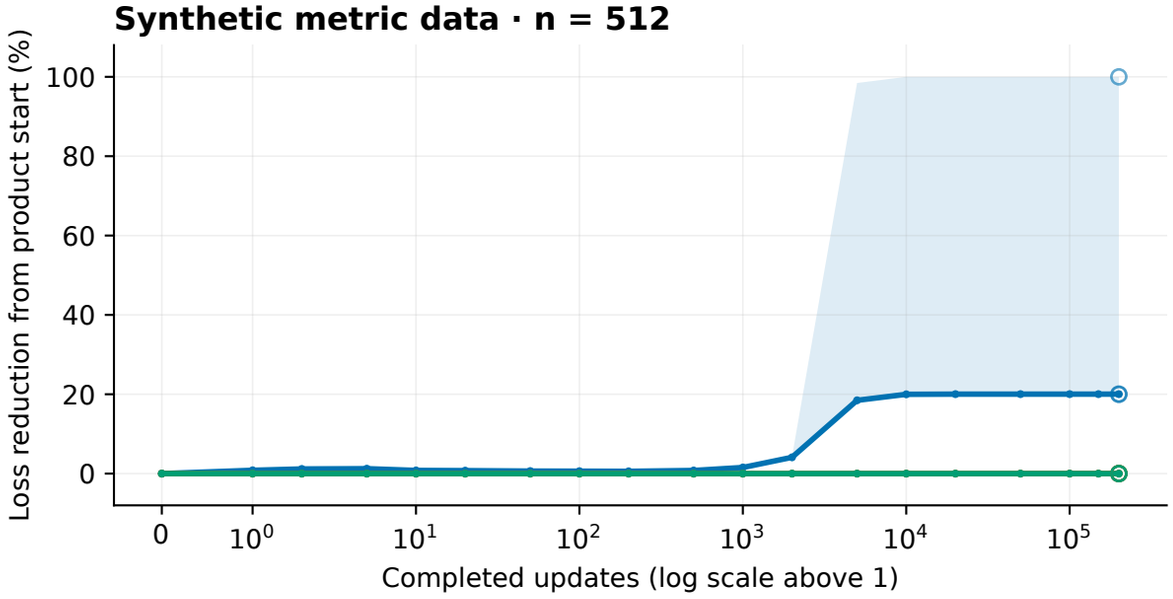
Pair-sampling comparison: relative loss reduction

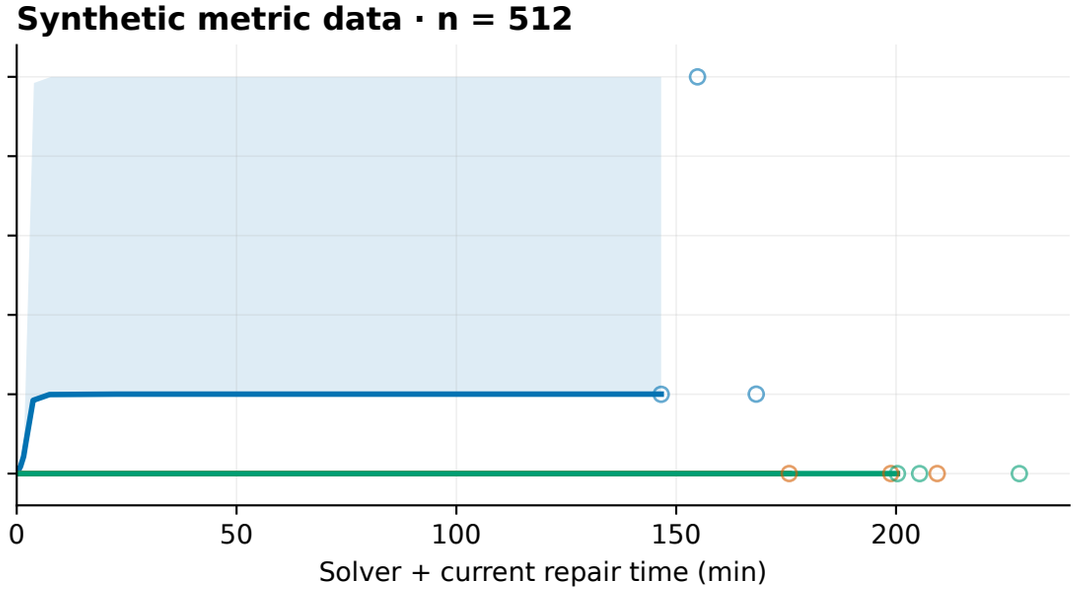
Preliminary · 3 stochastic seeds per method · 1 A100 per run · 1,024 pairs per gradient

FraMES · Entropic SMD (fresh gradient) · MSBPG (theorem parameters)

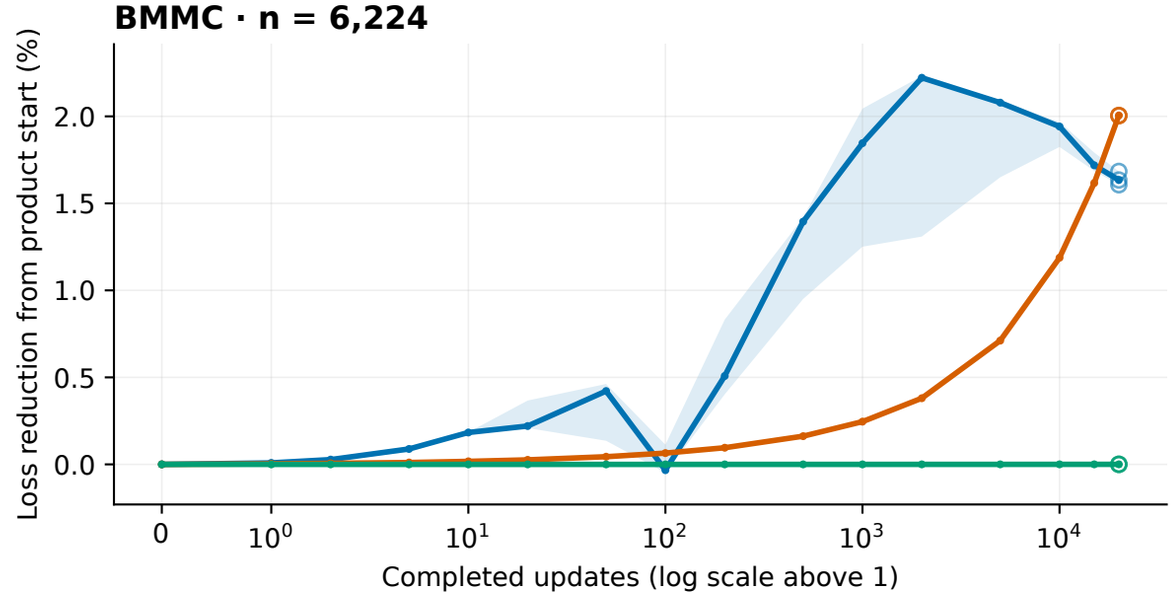
Synthetic metric data · n = 512



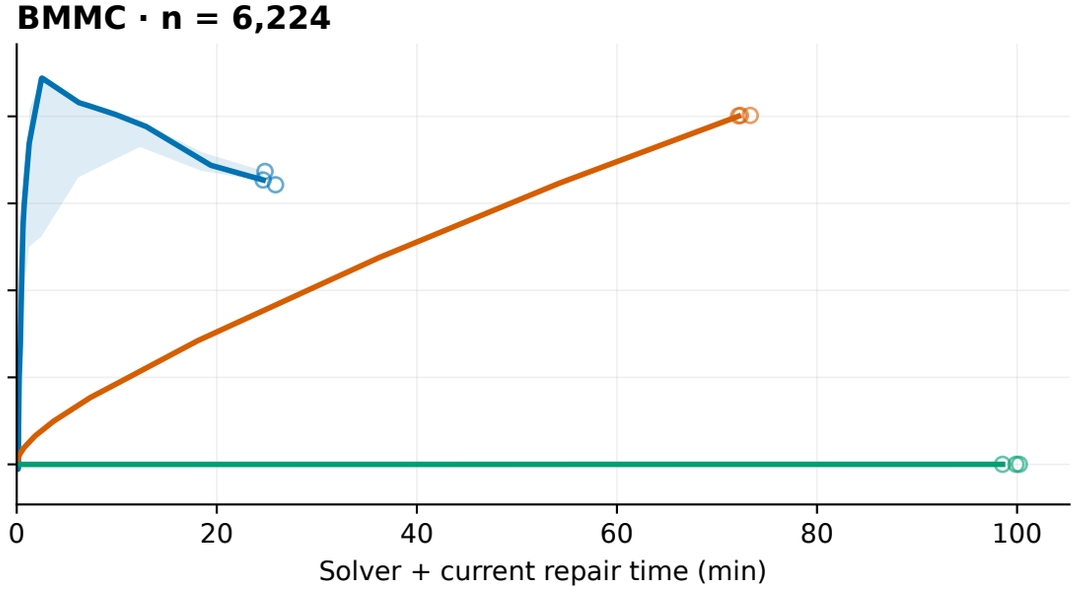
Synthetic metric data · n = 512



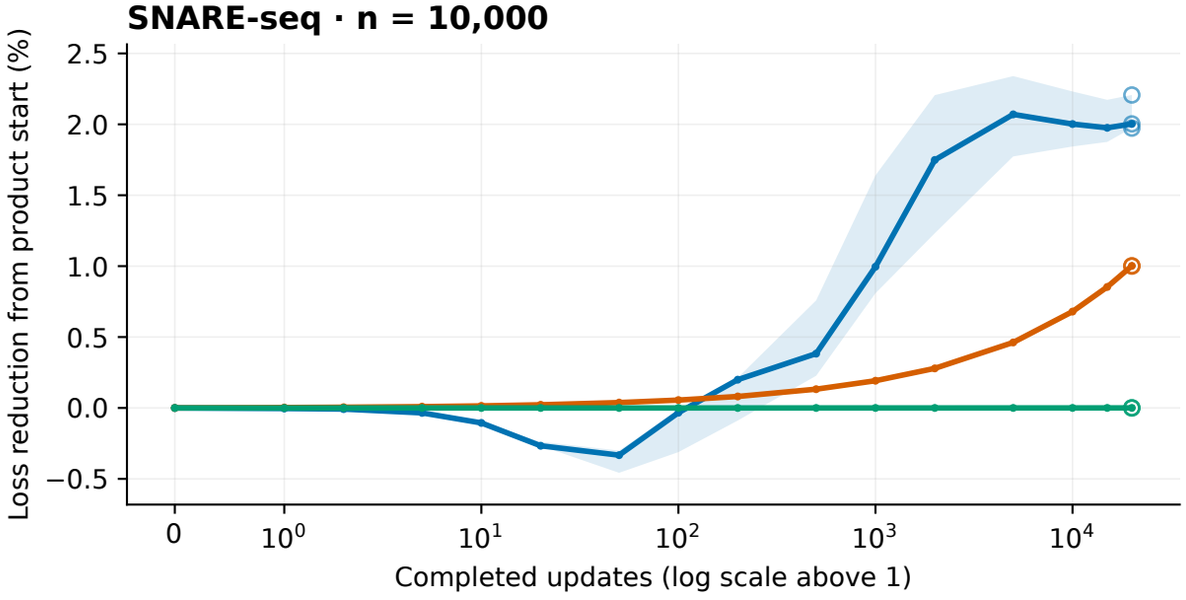
BMMC · n = 6,224



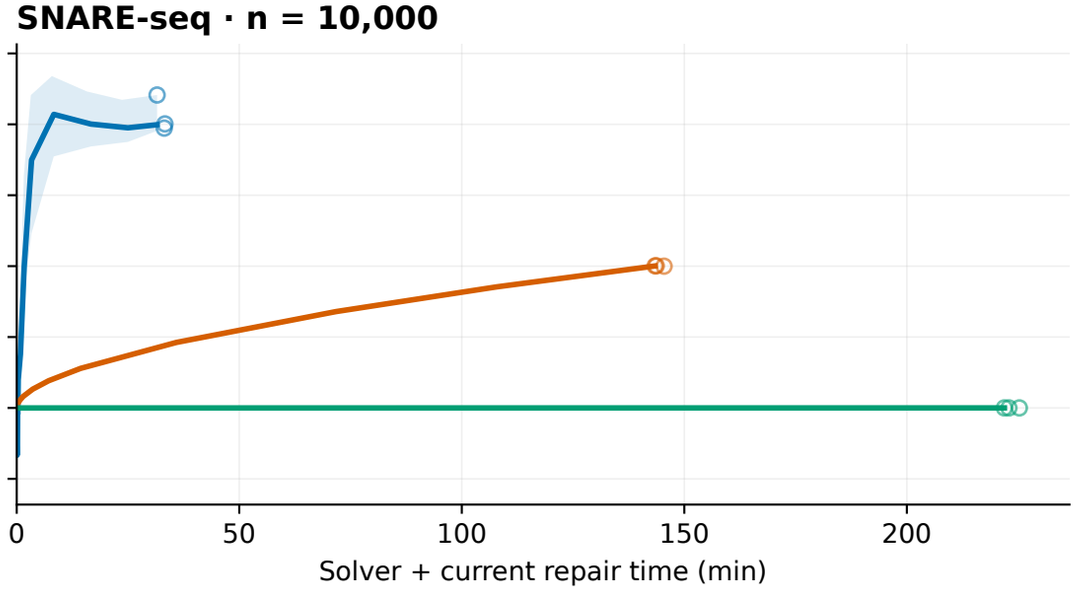
BMMC · n = 6,224



SNARE-seq · n = 10,000



SNARE-seq · n = 10,000



Lines: median; shaded bands: full range over three seeds (not confidence intervals). Hollow circles: independently reevaluated FP64 final outputs.
Time curves linearly interpolate logged checkpoints within each method's three-seed overlap; sparse logs do not resolve within-interval changes.
Time excludes diagnostics, logging, preparation and final validation. All curves use current outputs; no best-so-far selection. SHARE-seq is excluded.
Reduction = 100 × (initial online loss − current loss) / initial online loss, separately for each seed.